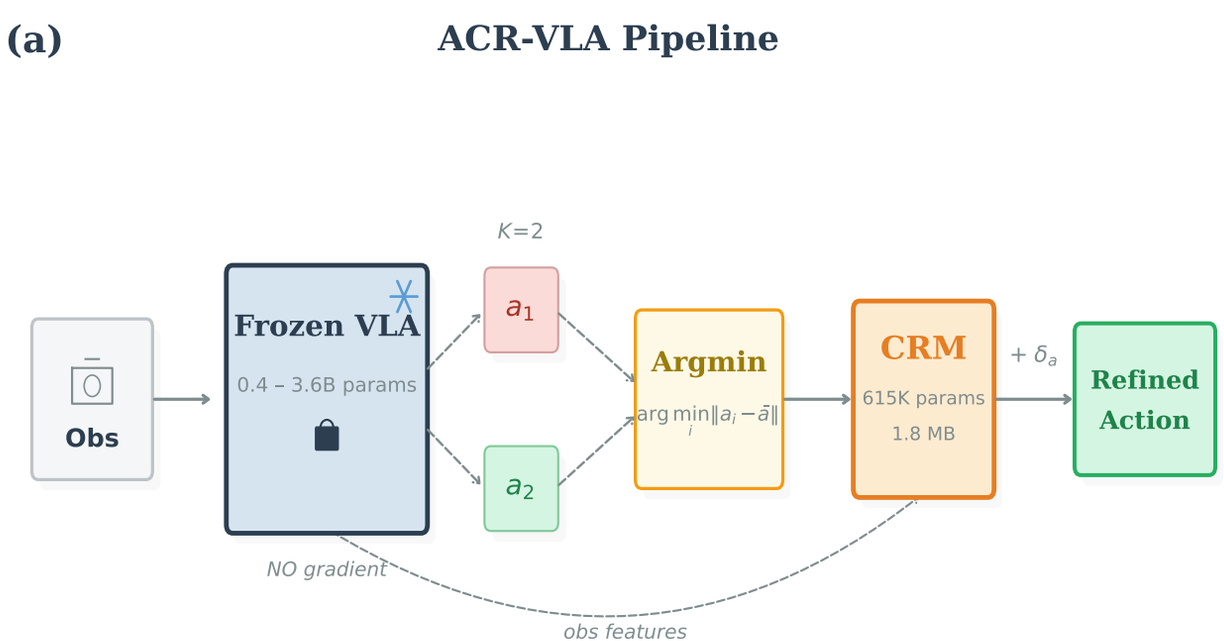
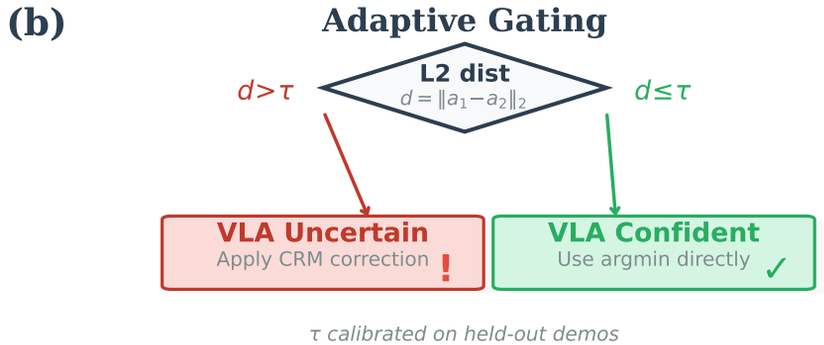
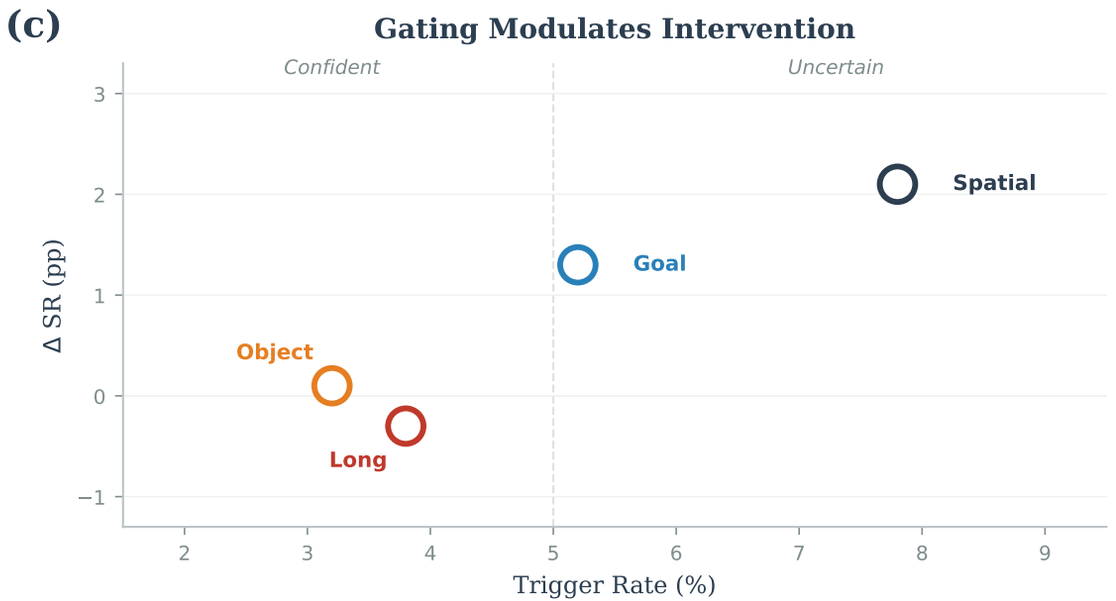

(a)
ACR-VLA Pipeline
Obs
Frozen VLA
0.4 – 3.6B params
K=2
a_1
a_2
Argmin
$\arg\min_i \|a_i - \bar{a}\|$
CRM
615K params
1.8 MB
$+ \delta_a$
Refined Action
NO gradient
obs features


(b)
Adaptive Gating
L2 dist
$d = \|a_1 - a_2\|_2$
$d > \tau$
$d \leq \tau$
VLA Uncertain
Apply CRM correction
VLA Confident
Use argmin directly
τ calibrated on held-out demos


(c)
Gating Modulates Intervention
Confident
Uncertain
Spatial
Goal
Object
Long
Δ SR (pp)
Trigger Rate (%)
3
2
1
0
−1
2
3
4
5
6
7
8
9